Downforce Optimization of an Airfoil in Ground Effect

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Application and model. We will maximize downforce on a racing-car front-wing section under drag and geometric limits. The initial NACA 0012 has reference chord $c = 0.50\,\mathrm{m}$, clearance $0.05\,\mathrm{m}$, and incidence $-3°$; incidence will vary. Speed is $200\,\mathrm{km/h}$ ($M \approx 0.163$, $Re_c \approx 1.90 \times 10^6$). The open-source MACH-Aero framework provides ADflow steady RANS on a quasi-2D overset O-grid and background grid. We plan a moving no-slip ground, $\boldsymbol{U}_{\mathrm{wall}} = (U_\infty, 0, 0)$, excluding finite-span and vehicle effects.

Model evaluation. Inputs are FFD shape, incidence, and operating conditions; outputs are C_L, C_D, pressure, thickness, and clearance. A preliminary 88,548-cell slip-ground solve took 150 s on four MPI processes; moving-wall and adjoint costs are unmeasured. We will use pyGeo FFD, IDWarp mesh deformation, and ADflow discrete-adjoint derivatives, and perform a grid-independence study.

Optimization problem. Let $x = (d)$ contain independent FFD displacements and incidence; n depends on FFD resolution and coupling. With upward-positive lift and flow state $u(x)$,

$$
\begin{aligned}
\text{minimize} \quad & f(x) = C_L(x, u(x)) \\
\text{by varying} \quad & x = (d) \in \mathbb{R}^n \\
\text{subject to} \quad & C_D(x, u(x))/C_{D,\max} - 1 \le 0, \\
& 1 - t_j(x)/t_{\min,j} \le 0, \quad j = 1, \ldots, N_t, \\
& X_{\min} - X_k(x) \le 0, \quad X_k(x) - X_{\max} \le 0, \\
& h_{\min} - Y_k(x) \le 0, \quad Y_k(x) - h_{\min} - H \le 0, \\
& r(u(x), x) = 0.
\end{aligned}
$$

Here $r = 0$ denotes the RANS residual. The box inequalities apply to all N_s surface points (X_k, Y_k). Subscript 0 denotes the baseline. Box containment and minimum thickness are the only geometric constraints. This is a continuous, constrained, uni-model, nonlinear program, solved in reduced space.

Software and expected learning. We plan to try IPOPT (interior point) and SLSQP (SQP) through pyOptSparse. The main expected learning is to get an intuition about the performance gain from geometry change. We will also conduct a trade study by relaxing $C_{D,\max}$ (Drag-limit sweep) and enlarging the design box. These scenarios represent a vehicle design that permits more front-wing drag or regulations that allow a larger wing envelope. We will try to quantify the additional downforce available under these situations and identify which changes in wing shape and incidence produce the gains. Pressure distributions and flow behavior will help explain these improvements and reveal where further relaxation provides diminishing returns.

Project goals / problem sequence

Stage	Investigation / deliverable
P0	Baseline, h/c sweep, grid independence.
P1	Incidence/ride height; contours, optimizer paths.
P2	**Minimum: single-element shape and incidence optimization.**
P3	Drag-limit sweep; other starting shapes.
P4	Vary box height, length, and clearance.
P5	Main element plus flap.
P6	Wake pressure loss as second objective.
P7	Compare one, two, and three elements.

P3–P7 are stretch goals.

